

Generalized wording for partial specializations

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Revision History

Revision 2 - 2020-08-17

- Incorporated additional review feedback from CWG telecon.

Revision 1 - 2020-03-23

- Incorporated review feedback from CWG telecon.

Revision 0 - 2020-02-01

- Initial draft.

Introduction

This paper is intended to address CWG issue [1711](#) by generalizing the existing wording for class template partial specializations to allow for partial specializations of variable templates.

Wording

All modifications are presented relative to [N4849](#). "[...]" indicates elided content that is to remain unchanged.

Modify §9.9 [\[namespace.udcl\]](#) paragraph 11:

[*Note*: Partial specializations of ~~class~~ templates are found by looking up the primary ~~class~~ template and then considering all partial specializations of that template. If a *using-declaration* names a ~~class~~ template, partial specializations introduced after the *using-declaration* are effectively visible because the primary template is visible ([\[temp.class.spec\]](#)). — *end note*]

Modify §13.1 [\[temp.pre\]](#) paragraph 4:

A *template-declaration* can appear only as a namespace scope or class scope declaration. Its *declaration* shall not be an *export-declaration*. In a function template declaration, the last component of the *declarator-id* shall not be a *template-id*. [*Note:* That last component may be an *identifier*, an *operator-function-id*, a *conversion-function-id*, or a *literal-operator-id*. In a class *or variable* template declaration, if the *class-declared* name is a *simple-template-id*, the declaration declares a *class-template* partial specialization ([\[temp.class.spec\]](#)). — *end note*]

Modify §13.1 [\[temp.pre\]](#) paragraph 6:

A template name has linkage ([\[basic.link\]](#)). Specializations (explicit or implicit) of a template that has internal linkage are distinct from all specializations in other translation units. A template, an *explicit specialization* ([\[temp.expl.spec\]](#)), and a *class-template* partial specialization shall not have C linkage. Use of a linkage specification other than "C" or "C++" with any of these constructs is conditionally-supported, with implementation-defined semantics. Template definitions shall obey the one-definition rule ([\[basic.def.odr\]](#)). [*Note:* Default arguments for function templates and for member functions of class templates are considered definitions for the purpose of template instantiation ([\[temp.decls\]](#)) and must also obey the one-definition rule. — *end note*]

Modify §13.4.3 [\[temp.arg.template\]](#) paragraph 1:

A *template-argument* for a template *template-parameter* shall be the name of a class template or an alias template, expressed as *id-expression*. ~~When the *template-argument* names a class template, o~~ Only primary *class* templates are considered when matching the template *template-argument* with the corresponding parameter; partial specializations are not considered even if their parameter lists match that of the template *template-parameter*.

Modify §13.4.3 [\[temp.arg.template\]](#) paragraph 2:

Any partial specializations ([\[temp.class.spec\]](#)) associated with the primary *class* template ~~or primary variable template~~ are considered when a specialization based on the template *template-parameter* is instantiated. If a specialization is not visible at the point of instantiation, and it would have been selected had it been visible, the program is ill-formed, no diagnostic required. [...]

Modify §13.7 [\[temp.decls\]](#) paragraph 1:

[*Note:* A *template-id*, that is, the *template-name* followed by a *template-argument-list* shall not ~~cannot~~ be specified in the declaration of the declared name of a primary template declaration.

[*Example:*

```
template<class T1, class T2, int I> class A<T1, T2, I> { }; // error  
template<class T1, int I> void sort<T1, I>(T1 data[I]); // error
```

— *end example*]

[~~Note:~~ However, this syntax is allowed in ~~class template~~ partial specializations ([temp.class.spec]). — end note]

Rename §13.7.5 [temp.class.spec]:

Class template partial specializations

Modify §13.7.5 [temp.class.spec] paragraph 1:

A ~~primary-class~~ template declaration is one in which the ~~class template~~ name of the template is an identifier not followed by a template-argument-list. A template declaration in which the ~~class~~ template name is a *simple-template-id* is a *partial specialization* of the ~~class~~ template named in the *simple-template-id*, which shall be a class or variable template. A partial specialization of a ~~class~~ template provides an alternative definition of the template that is used instead of the primary definition when the arguments in a specialization match those given in the partial specialization ([temp.class.spec.match]). The primary template shall be declared before any partial specializations of that template. A partial specialization shall be declared before the first use of a ~~class~~ template specialization that would make use of the partial specialization as the result of an implicit or explicit instantiation in every translation unit in which such a use occurs; no diagnostic is required.

Modify §13.7.5 [temp.class.spec] paragraph 2:

Each ~~class template~~ partial specialization is a distinct template ~~and definitions shall be provided for the members of a template partial specialization~~ ([temp.class.spec.mfunc]).

Modify §13.7.5 [temp.class.spec] paragraph 4:

A ~~class template~~ partial specialization may be constrained ([temp.pre][temp.constr]). [...]

Modify §13.7.5 [temp.class.spec] paragraph 5:

The template parameters of a template are specified in the angle bracket enclosed list that immediately follows the keyword `template`. For partial specializations, the template argument list is explicitly written immediately following the ~~class~~ template name. For primary templates, this list is implicitly described by the template parameter list ([temp.dep.type]). Specifically, the order of the template arguments is the sequence in which they appear in the template parameter list. [Example: The template argument list for the primary template in the example above is `<T1, T2, I>`. — end example] [Note: The template argument list cannot be specified in the primary template declaration. For example,

```
template<class T1, class T2, int I>
class A<T1, T2, I> { }; // error
```

— end note]

Modify §13.7.5 [\[temp.class.spec\]](#) paragraph 6:

A ~~class template~~ partial specialization may be declared in any scope in which the corresponding primary template may be defined ([\[namespace.memdef\]](#), [\[class.mem\]](#), [\[temp.mem\]](#)). [...]

Modify §13.7.5 [\[temp.class.spec\]](#) paragraph 7:

Partial specialization declarations themselves are not found by name lookup. Rather, when the primary template name is used, any previously-declared partial specializations of the primary template are also considered. [*Note*: One consequence is that a *using-declaration* which refers to a ~~class~~ template does not restrict the set of partial specializations which may be found through the *using-declaration*. — *end note*] [...]

Modify §13.7.5 [\[temp.class.spec\]](#) paragraph 9:

Within the argument list of a ~~class template~~ partial specialization, the following restrictions apply:

- The type of a template parameter corresponding to a specialized non-type argument shall not be dependent on a parameter of the [\[partial\]](#) specialization. [*Example*:

```
template <class T, T t> struct C {};
template <class T> struct C<T, 1>; // error
template< int X, int (*array_ptr)[X] > class A {};
int array[5];
template< int x > class A<x,&array> { }; // error
```

— *end example*]

- The [\[partial\]](#) specialization shall be more specialized than the primary template ([\[temp.class.order\]](#)).
- The template parameter list of a [\[partial\]](#) specialization shall not contain default template argument values.
- An argument shall not contain an unexpanded pack. If an argument is a pack expansion ([\[temp.variadic\]](#)), it shall be the last argument in the template argument list.

Rename §13.7.5.1 [\[temp.class.spec.match\]](#):

Matching of ~~class template~~ partial specializations

Modify §13.7.5.1 [\[temp.class.spec.match\]](#) paragraph 1:

When a ~~class~~ template is used in a context that requires an instantiation of the ~~class template~~, it is necessary to determine whether the instantiation is to be generated using the primary template or one of the partial specializations. This is done by matching the template arguments of the ~~class~~ template specialization with the template argument lists of the partial specializations.

- If exactly one matching [\[partial\]](#) specialization is found, the instantiation is generated from that [\[partial\]](#) specialization.

- If more than one matching partial specialization is found, the partial order rules ([\[temp.class.order\]](#)) are used to determine whether one of the partial specializations is more specialized than the others. If ~~none of the specializations is more specialized than all of the other matching specializations~~ such a partial specialization exists, the instantiation is generated from that partial specialization; otherwise, ~~then~~ the use of the ~~class~~ template is ambiguous and the program is ill-formed.
- If no matches are found, the instantiation is generated from the primary template.

Modify §13.7.5.1 [\[temp.class.spec.match\]](#) paragraph 4:

In a type name that refers to a class template specialization, of a class or variable template (e.g., `A<int, int, 1>`), the argument list shall match the template parameter list of the primary template. The template arguments of a partial specialization are deduced from the arguments of the primary template.

Rename §13.7.5.2 [\[temp.class.order\]](#):

Partial ordering of class template partial specializations

Modify §13.7.5.2 [\[temp.class.order\]](#) paragraph 1:

For two class template partial specializations, the first is *more specialized* than the second if, given the following rewrite to two function templates, the first function template is more specialized than the second according to the ordering rules for function templates ([\[temp.func.order\]](#)):

- Each of the two function templates has the same template parameters and associated constraints ([\[temp.constr.decl\]](#)) as the corresponding partial specialization.
- Each function template has a single function parameter whose type is a class template specialization where the template arguments are the corresponding template parameters from the function template for each template argument in the *template-argument-list* of the *simple-template-id* of the partial specialization.

[...]

Rename §13.7.5.3 [\[temp.class.spec.mfunc\]](#):

Members of class template partial specializations

Modify §13.7.5.3 [\[temp.class.spec.mfunc\]](#) paragraph 1:

The template parameter list of a member of a class template partial specialization shall match the template parameter list of the class template partial specialization. The template argument list of a member of a class template partial specialization shall match the template argument list of the class template partial specialization. ~~A class template partial specialization is a distinct template.~~ The members of the class template partial specialization are unrelated to the members of the

primary template. Class template partial specialization members that are used in a way that requires a definition shall be defined; the definitions of members of the primary template are never used as definitions for members of a class template partial specialization. An explicit specialization of a member of a class template partial specialization is declared in the same way as an explicit specialization of a member of the primary template. [...]

Modify §13.7.8 [\[temp.concept\]](#) paragraph 5:

A concept is not instantiated ([\[temp.spec\]](#)). [*Note*: A concept-id ([\[temp.names\]](#)) is evaluated as an expression. A concept cannot be explicitly instantiated ([\[temp.explicit\]](#)), explicitly specialized ([\[temp.expl.spec\]](#)), or partially specialized ([\[temp.class.spec\]](#)). — *end note*]

Modify §13.8.2.1 [\[temp.dep.type\]](#) paragraph 1:

A name refers to the *current instantiation* if it is

- in the definition of a class template, a nested class of a class template, a member of a class template, or a member of a nested class of a class template, the injected-class-name ([\[class.pre\]](#)) of the class template or nested class,
- in the definition of a primary class template or a member of a primary class template, the name of the class template followed by the template argument list of the primary template (as described below) enclosed in <> (or an equivalent template alias specialization),
- in the definition of a nested class of a class template, the name of the nested class referenced as a member of the current instantiation, or
- in the definition of a class template partial specialization or a member of a class template partial specialization, the name of the class template followed by the template argument list of the partial specialization enclosed in <> (or an equivalent template alias specialization). If the n^{th} template parameter is a template parameter pack, the n^{th} template argument is a pack expansion ([\[temp.variadic\]](#)) whose pattern is the name of the template parameter pack.

Optional: Rename the following stable names as shown:

Old Name	New Name
temp.class.spec	temp.spec.partial
temp.class.spec.match	temp.spec.partial.match
temp.class.order	temp.spec.partial.order
temp.class.spec.mfunc	temp.spec.partial.member

Acknowledgments

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References

1. CWG issue 1711: <http://wg21.link/cwg1711>